

TenD Self-Dual Five-Form Project Audit

Document title: TenD_SelfDual_FiveForm_Project_Audit

Audit date: 2026-08-06

Auditor role: forensic technical audit (documentation and provenance only)

Nondestructive: no existing project files were modified, renamed, moved, overwritten, or deleted.

Outputs directory: Project_Audit_2026-08-06/

Scopes inspected:

1. Mathematica Research folder: /Users/davidrabinow/Downloads/26-27 Prep/Stux Technologies/Research/

2. Python MIT repository: /Users/davidrabinow/Projects/UPRSxxXCEL/Research/scalar-invariants-of-tensors/MIT/

Primary papers:

- Elamaran-Ferko-Scarlett, Machine Learning Invariants of Tensors, arXiv:2512.23750 (local PDF present).

- Cederwall-Hutomo-Kuzenko-Lechner-Sorokin, Some remarks on invariants, arXiv:2509.14350 (cited in project notes; no local PDF found in Research folder).

- Hutomo-Lechner-Sorokin, On non-linear chiral 4-form theories in D=10, arXiv:2509.14351 (cited; no local PDF found).

Mathematica availability to auditor: wolframscript / math not found. Therefore Mathematica modules are classified OBSERVED RUN when RESEARCH_LOG / exported catalogs / TestReport design strongly indicate prior execution, not RERUN PASSED.

Python availability: smoke rerun performed 2026-08-06 (graphs N=2,4,6; blind discovery through degree 6; self-duality). Result: SMOKE_OK (see supporting/rerun_smoke.txt).

1. Executive summary

Ultimate research question

What are the independent Lorentz-scalar polynomial invariants of a real self-dual (chiral) 5-form ($F=F$) in ten-dimensional Lorentzian spacetime, what are their degrees and generating structure, and what are the syzygies among candidate contractions--starting from a validated reproduction of the six-dimensional Euclidean 3-form analysis of Elamaran-Ferko-Scarlett?

Strongest current 6D result

Strongest claim that is well supported: the project has reproduced, in exact Mathematica arithmetic (Gates 1-5, as recorded in RESEARCH_LOGv4.md) and independently in a Python blind discovery path (tensor_invariants / outputs/6d/), the paper's connected graph counts $((1,2,6,20))$, same-degree ranks $((1,2,3,6))$, and generator-degree pattern $([2,4,4,6,8])$ for metric-only contractions of a generic antisymmetric 3-form in Euclidean ($d=6$). Exact Jacobian rank 5 for the five paper polynomials and exact vanishing of Appendix A relations on finite integer holdouts are recorded for Mathematica. Python discovery ranks are primarily SVD with modular syzygy checks.

Strongest current 10D result

Strongest claim that is well supported in code artifacts: Lorentzian self-duality conventions with ($\wedge^2=+1$), 252 generic / 126 chiral components; degree-4 recovery of a 1-dimensional invariant space identified with ($I_4=\text{tr } M^2$); degree-6 census (12043) labeled / (54) canonical / (49) connected with two independent directions (trace and ($N^{(1050)}$)); degree-8 Burnside census (1753) canonical ((1689) connected) with a seven-dimensional validated basis including ($I_4\wedge^2$) and six named graphs; degree-10 foundations only validating the two lower products and stating a conditional target of 12 new directions. These 10D computational claims are OBSERVED RUN from Mathematica sources/exports; they were not rerun by this auditor. The Python 10D path has no generators.

Maximum degree investigated

- 6D: through degree 8 exactly (graphs/ranks); Appendix relations at 6 and 8; paper stabilization through 18 is cited but not shown as executed in RESEARCH_LOG Stages 1-5. Python checked ($n_{\text{new}}=0$) at degree 10 on a sample of 12 graphs.
- 10D: foundations through degree 10 products; graph discovery through degree 8; no degree-10 graph enumeration completed.

What has / has not been established

Established (with caveats)	Not established
--- ---	
6D paper graph counts & ranks (exact MMA; Python match)	Completeness of 6D five-generator set as full invariant ring
6D five polys algebraically independent (exact Jacobian)	Symbolic proof of Appendix A identities
10D conventions / self-duality	Independent proof of Krull dimension 81
10D low-degree Hilbert targets copied/checked as integers	That metric-only graphs exhaust ($SO(1,9)$) invariants
Degree-8 seven-direction finite-sample certificate (claimed)	Reduction of all 1753 degree-8 graphs; degree-10 generators
Full syzygy resolution; Hironaka decomposition; 81-parameter structure	

Most important methodological risk

Matching a published singlet dimension (e.g. 7 at degree 8) using metric-only graph evaluation does not by itself prove that every ($SO(1,9)$) scalar built with Levi-Civita tensors reduces to the metric-only span after ($F=F$). Without an explicit epsilon-reduction certificate for the project's conventions, Hilbert agreement is conditional evidence of completeness, not a proof. Secondary risks: Mathematica kernel-state contamination across Get chains; missing TenDDegree6ContractionPlanning_V1.wl; Python 6D discovery using SVD as primary truth; degree-10 claims easily overstated as "12 generators found" when only product rank 2 was validated.

Estimated completion ranges (not mathematical measurements)

Workstream	Estimated range	Rationale
--- --- ---		
6D reproduction vs arXiv:2512.23750 ?4.1 + App. A	85-95%	Counts, ranks, independence, App. A holdouts done; stabilization through 18 and symbolic identities incomplete

| 10D low-degree study (deg <=8 vs Cederwall targets) | 55-70% | Deg 4/6/8 substantial
Mathematica work + catalogs; not auditor-rerun; epsilon/S0 gap; missing planning file |
| Complete ultimate project (81-parameter generators+syzygies) | 5-15% | Only low degrees; no
deg-10 discovery; no full syzygies; no completeness proof |

2. Complete file inventory

Machine-readable full inventory: TenD_Project_File_Manifest.csv (204 rows + missing
dependency). SHA-256 digests: supporting/sha256_all_research.txt,
supporting/sha256_mit_core.txt.

2.1 Mathematica modules (current recommended chain)

Use these (latest):

AntisymmetricPForms.wl -> MetricContractions6D.wl -> FunctionalIndependence6D.wl ->
AppendixRelations6D.wl -> GraphEnumeration6D_FIXED.wl -> InvariantBenchmarks_CORRECTED_V2.wl
-> TenDLorentzianFoundations_V2.wl -> TenDRepresentationTargets.wl ->
TenDDegree4Invariants_V4.wl -> TenDDegree6TraceInvariant_V1.wl +
TenDDegree6N1050Invariant_V2.wl + TenDDegree6GraphEnumeration_V1.wl + TenDDegree6Catalog_V2.wl
-> TenDDegree8GraphEnumeration_V1.wl + Degree8CanonicalGraphKeys.wl + discovery batches +
TenDDegree8BasisValidation_V1.wl + TenDDegree8FormulaCatalog_V1.wl ->
TenDDegree10Foundations_V1.wl.

Do not use (obsolete / superseded):

GraphEnumeration6D.wl; InvariantBenchmarks.wl; TenDLorentzianFoundations_V1.wl;
TenDDegree4Invariants_V1-V3.wl; TenDDegree6N1050Invariant_V1.wl;
TenDDegree6GraphBasis_FinalK6_V1.wl; TenDDegree6Catalog_V1.wl (wrong K6 sign); duplicate
RESEARCH_LOG copies (byte-identical).

Missing but required by Batch A: TenDDegree6ContractionPlanning_V1.wl -- ABSENT. Status: NOT
RUN / broken dependency for clean-kernel Batch A.

Named in audit brief but absent: MetricContractions6D_FIXED.wl, MetricContractions6D_FINAL.wl,
TensorContractionOrderingFix.wl (axis fix absorbed into MetricContractions6D.wl / Gate 3
narrative).

2.2 Notebook

6D_3Form_Project.nb: Mathematica notebook (Wolfram 15.0). Heuristic parse: 6 Cell[markers; 0
VerificationTest strings; not a reliable stored TestReport archive. Status: UNKNOWN for
scientific outputs. Do not infer Gate results from this notebook alone.

2.3 Python MIT package

Primary science: tensor_invariants/, run_pipeline.py, outputs/6d/, outputs/10d/, reports/.
Legacy (ops / superseded for discovery): src/invariants/, src/invariant_engine/,
scripts/run_6d.py.

Do not treat docs/invariants_78.json as validated 10D science.

3. Chronological research log

```
| When | Event | Outcome |
|---|---|---|
| 2026-08-03 | Stage 1: AntisymmetricPForms | 5/5 exact tests OBSERVED |
| 2026-08-03 | Stage 2: MetricContractions; AssociationMap bug | Fixed to Map[evaluate,...];
5/5 then 8 substantive True |
| 2026-08-03 | Stage 3: Jacobian; 0^0 Indeterminate; then rank-4 because (x^(6)) was (X_1^(6))
| Axis-order / RaiseTensorIndex fix; rank 5; A.2a residual 0 |
| 2026-08-03 | Stage 4: Appendix relations | 7/7; 12 integer holdouts seed 20260804 |
| 2026-08-03 | Stage 5: Graph enum; IncidenceGraph name collision | FIXED module; counts
1,2,6,20; ranks 1,2,3,6 |
| 2026-08-03 | Literature gate for 10D | Hilbert/Euler/81 sourced; computational 10D not yet |
| 2026-08-03 | Lorentzian V1->V2; Degree4 V1->V4 | Component keys; diagonal raise Dot;
Transpose Ordering |
| 2026-08-03->05 | Degree6 enum/trace/N1050/batches/catalog | 12043/54/49; K6 sign corrected
V2 |
| 2026-08-05->06 | Degree8 Burnside, keys, discovery A-D, basis validation, formula catalog |
1753; basis 7 directions; CSV/TXT export |
| 2026-08-06 01:00 | TenDDegree10Foundations_V1.wl | Product subspace module authored |
| 2026-08-06 ~01:12-09:07 | Python MIT package + pipeline | Blind 6D PASS outputs; 10D smoke
only |
```

RESEARCH_LOG.md, _2, v3, v4 are byte-identical (MD5 431f853d...) and stop at Stage 5 / Module 6 plan--they do not document 10D degree work. Provenance for 10D is filename versions + exported catalogs + module Print banners.

4. Reconstruction of the paper's method (arXiv:2512.23750 ??2-4)

Encoding and enumeration

Contractions of an antisymmetric (p)-form with the metric are encoded as loopless weighted multigraphs with vertex weighted degree (p) (for 3-forms: 3-regular). Edge weight = number of contracted index pairs. Self-loops are excluded by antisymmetry. The paper prefers connected graphs; disconnected graphs factorize into products (secondary / composite invariants).

Isomorphism and evaluation

Graphs are reduced up to isomorphism (paper: VF2-style). Random tensors are sampled; contractions evaluated; fixed-degree linear dependence is extracted from numerical evaluation matrices (paper: floating-point SVD nullspaces). New primary generators are those independent of polynomials in lower-degree invariants. Syzygies are reconstructed from null vectors and validated on fresh samples. Stabilization over several degrees is heuristic evidence, not a completeness proof (? paper discussion / project RESEARCH_LOG caution).

Project differences from the paper

```
| Paper | This project |
|---|---|
| Floating-point SVD discovery | Mathematica exact/modular ranks; Python SVD + modular
syzygies |
| igraph isomorphism | MMA CanonicalGraph incidence encoding; Python NetworkX / min-lex |
```

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5. Six-dimensional reproduction

Definitions (paper Eqs. (4.1)-(4.4); Mathematica MetricContractions6D.wl)

```
[
aligned
x^(2) &= H_abcH^abc,\\
x^(4)_1 &= H_abcH_adeH^defH^bc_f,\\
x^(4)_2 &= H_abcH_adeH^cefH^bd_f,\\
x^(6) &= H_abcH^chiH_ghiH^adgH_defH^bef,\\
x^(8) &= H_abcH^bciH_ghiH^gjkH_jklH^fhlH_defH^ade.
aligned
]
```

(Index raising with Euclidean ().) Mathematica stores contraction specs as paired index lists, e.g. "x^(2)" -> "a","b","c","a","b","c".

Counts vs paper ?4.1

Degree	Connected	graphs (paper)	Project MMA Gate5	Project Python
2	1	1	1	1
4	2	2	2	2
6	6	6	6	6
8	20	20	20	20

Degree	Connected	rank (paper)	MMA	Python
2	1	1	1	1
4	2	2	2	2
6	3	3	3	3
8	6	6	6	6

Degree	New gens	after products	Python blind
2	1	1	1
4	2	2	2
6	1	1	1
8	1	1	1
10	0 (paper)	0	on sample of 12

Final degrees: ([2,4,4,6,8]).

Jacobian / modular / Appendix

- Exact Jacobian (520) rank 5 over (Q) and primes (1000003,1000033,1000037) (Gate 3) -- REPRODUCED COMPUTATIONALLY WITH EXACT ARITHMETIC (OBSERVED RUN).
- Appendix (A.2a),(A.2b),(A.4),(A.6) exact zero on 12 holdouts (Gate 4) -- FINITE-SAMPLE EVIDENCE ONLY (exact arithmetic on samples != symbolic proof).

- Stabilization through degree 18: stated in paper; not executed in RESEARCH_LOG Stages 1-5.

Evidence labels for 6D headline

- Graph counts: REPRODUCED COMPUTATIONALLY WITH EXACT ARITHMETIC (MMA) + RERUN PASSED for $N \leq 6$ Python smoke.
- Algebraic independence of five polys: REPRODUCED COMPUTATIONALLY WITH EXACT ARITHMETIC (Jacobian) $\Rightarrow$ independence in characteristic zero at tested point; not completeness.
- Python blind (n_{new}): SUPPORTED ONLY BY FLOATING-POINT NUMERICS (SVD), cross-checked by tests.

6. Benchmarks

InvariantBenchmarks_CORRECTED_V2.wl (12 VerificationTests; OBSERVED RUN): one/two vectors under $(O(d))$; real antisymmetric 2-form in 4D under $(SO(4))$ and $(O(4))$.

Pfaffian: $(P=18_{\text{abcd}}F^{\text{ab}}F^{\text{cd}})$ is an $(SO(4))$ invariant. Under orientation-reversing $(O(4))$ elements, $()$ flips sign, so (P) is a pseudoscalar (changes sign) and is not an $(O(4))$ invariant. The corrected module's TestID explicitly distinguishes $(SO(4))$ from $(O(4))$. Earlier InvariantBenchmarks.wl had a missing Dot ($\text{DiagonalMatrix}[\dots] \text{Reverse}[\dots]$) and is SUPERSEDED*.

No separate self-dual 3-form benchmark module beyond this path was identified as current.

7. Ten-dimensional mathematical formulation

Conventions (project)

- Metric: $(=\text{diag}(-1, +1^9))$ (mostly plus).
- Indices: $(0, \dots, 9)$; time = 0.
- Orientation: $(_0123456789=+1)$.
- Hodge: $((F)_1_5=15!_1_5_1_5F^{_1_5})$.
- Chirality: $(F=F)$.

$(^2)$ formula

[
 $^2 = (-1)^p(D-p)+t$.
]

For $((D,p,t)=(10,5,1))$: $(p(D-p)+t=25+1=26)$ even $\Rightarrow (^2=+1) \Rightarrow$ real $()$ eigenspaces of dimension $(252/2=126)$.

Euclidean $(t=0)$: $(^2=-1) \Rightarrow$ eigenvalues (i) ; no nonzero real self-dual 5-form.

Group-theoretic facts

- $(SO(1,9)=45)$.
- $(126-45=81)$ equals the Krull dimension only if the generic stabilizer is trivial -- STATED IN A CITED PRIMARY SOURCE (Cederwall et al.), UNRESOLVED as an independent project proof.
- Orientation reversal in $(O(1,9))$ exchanges chiralities $\Rightarrow$ fixed-chirality problem is naturally $(SO(1,9))$, not full $(O(1,9))$.
- Quadratic $(F_1_5F^{_1_5}=0)$ for self-dual odd middle forms -- ESTABLISHED ANALYTICALLY (and cited).

Distinctions

Chiral 4-form potential (A_4) != field strength ($F_5=dA_4$) != generic 5-form != self-dual projection. Invariant Lagrangians without derivatives are polynomials in (F_5).

8. Representation-theory targets

From TenDChiralFiveForm_Representation_Literature_Gate.md and TenDRepresentationTargets.wl (checks series conversion; does not re-run LiE):

Hilbert (Cederwall Eq. (4.2)):

$(1 + t^4 + 2t^6 + 7t^8 + 14t^{10} + 72t^{12} + \dots)$

Euler/plethystic initial balances: $((m_4,)=(1,2,6,12,62,))$.

Source caveats: counts are for the complexified chiral (126) of $(SO(10,C))$ (Dynkin (00002)). Identifying them with real $(SO(1,9))$ metric-only graph spans requires additional hypotheses (complexification, epsilon reduction). Positive Euler coefficients are net generator-relation balances, not automatically raw generator counts. Sum of first five positive Euler exponents ($=83>81$) => relations must exist if the cited data hold.

9. Degree-by-degree 10D results

Degree	Published singlets	Labeled	Canonical	Connected	Exact span rank (claimed)	Lower product rank	New quotient	Status	
2	0	--	--	0 (vanishes)	0	0	Analytic + conventions		
4	1	21	5	4	1	0	1	OBSERVED RUN V4	
6	2	12043	54	49 (+5 disc.)	2	0	2	OBSERVED RUN; Catalog_V2	
8	7	45163496	1753	1689 (+64 disc.)	7	1	$((I_4^2))$	6	OBSERVED RUN; basis validation
10	14	--	--	--	products rank 2 claimed	2	target 12	Foundations only; no graph discovery	

Degree 4

Invariant $(I_4=\text{tr } M^2)$ with $(M^{\wedge} = F_{1\,2\,3\,4}F^{\wedge}_{1\,2\,3\,4})$ (factorial normalizations as in formula catalog). Graph+I4 span rank 1 claimed.

Degree 6

Two independents: $(I_6^{(1)}=\text{tr } M^3)$ and $(I_6^{(2)})$ from $(N^{\wedge}(1050))$ cubic (Hutomo/Cederwall). Catalog CSV: 54 graphs; K6 coefficients corrected to $(\backslash -17/1152, \backslash 125/4 \backslash)$ in V2. Some graph relations fitted on samples with holdout residuals recorded--not symbolic proofs.

Degree 8 -- seven validated basis directions

1. (I_4^2)
2. Graph 3
3. Graph 249
4. Graph 508
5. Graph 61

6. Graph 376
7. Graph 528

Explicit keys, adjacency matrices, and Einstein/metric contractions:

TenDDegree8InvariantFormulaCatalog.txt (copied to supporting/). Claimed: exact & modular rank 7 on 24 fresh tensors and on each 8-sample block. Not all 1753 graphs were necessarily evaluated for reduction; discovery used batched peak-rank/work filters. Matching published dimension 7 is conditional completeness evidence.

Degree 10

TenDDegree10Foundations_V1.wl is designed to run 16 VerificationTests validating products ($I_4 I_6^{\wedge}(1)$) and ($I_4 I_6^{\wedge}(2)$) have rank 2, hence conditional new-direction target 12 given published dim 14. This auditor did not observe a stored TestReport output file proving it was run; classify as SOURCE EXISTS / designed OBSERVED RUN unknown. Do not claim twelve degree-10 generators have been found.

Python explore-10d: only $N=2$ (1 graph) and $N=4$ (4 graphs) 5-regular censuses + self-duality.

10. S0 versus 0 and epsilon-tensor audit

```
| Layer | Status |
|---|---|
| Analytic: even number of ( ) reduces to metrics | ESTABLISHED ANALYTICALLY (standard) |
| Analytic: saturating one ( ) on five indices of self-dual (F) replaces by metric form |
ESTABLISHED ANALYTICALLY in saturated case; Cederwall App. C |
| Project computational proof that every ( $S_0(1,9)$ ) scalar with ( ) reduces to metric-only
graphs under project conventions | UNRESOLVED |
| Hilbert match at deg 8 => completeness without epsilon audit | Unsafe; requires hypotheses
that (i) cited singlets apply to the same real representation & allowed tensors, and (ii)
epsilon sectors add nothing new after ( $F=F$ ) |
```

Metric-only enumeration might omit orientation-sensitive invariants if reduction fails. Fixed chirality is an (S_0) problem; Pfaffian-like lessons from 4D benchmarks apply.

11. Graph canonicalization and antisymmetry-sign audit

- Topology determines a contraction only up to sign and slot wiring; modules assign fixed vertex index labels (see formula catalog).
- Canonicalization via incidence multigraphs aims to preserve multiplicities; whether all odd automorphisms forcing vanishing were proved symbolically vs detected on samples is not uniformly documented--many vanishings are sample/fit based (Catalog flags).
- Degree8CanonicalGraphKeys.wl SHA-256 proves file identity, not mathematical completeness of the orbit list.
- Python $N \leq 8$ uses exact min-lex labels; $N > 8$ uses fingerprints (weaker).

12. Claims register

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See TenD_Project_Claims_Register.csv (18 major claims with evidence labels).

13. Test register

See TenD_Project_Test_Register.csv.

Critical: a green TestReport after loading many files into one kernel may depend on leftover DownValues. Modules should be retested in a fresh kernel with documented Get order. Auditor could not execute Wolfram.

14. Failures and discrepancies

Symptom	Root cause	Corrected by	Downstream regenerated?	
---	---	---	---	
Module2 AssociationMap failures	Rule-wise Map over Association	Map[evaluate, specs]	in	
MetricContractions	Yes (Gate2 pass)			
Jacobian Indeterminate	(0^0)	Explicit degree-0 row = 1	Yes	
Jacobian rank 4	Wrong (x^6) (= (X_1^6))	axis order	TensorContract axis fix;	
RaiseTensorIndex	Yes; Gate3 rank5			
GraphEnumeration load fail	IncidenceGraph protected	_FIXED rename	Yes Gate5	
Benchmarks matrix bug	Missing Dot	_CORRECTED_V2	Yes	
Lorentzian missing components	Incomplete association keys	V2 Lookup fill	Yes	
Degree4 wrong raise/transpose	Diagonal raise; Ordering	V3/V4	Yes	
Degree6 K6 wrong sign	Catalog V1	Catalog V2 -17/1152	Catalog regenerated; confirm	
batches refreshed				
Missing ContractionPlanning	File absent	Unresolved	Batch A cannot clean-start	
Python initial audit: ladder omitted (P_N)	Design bug	tensor_invariants product quotient		
Yes 2026-08-06				
Legacy docs 78/81	Autonomous climb	Not authoritative	Do not use	

15. Reproducibility audit

Item	Finding	
---	---	
Required Mathematica	Logs cite Wolfram 15.0.1	
Auditor OS	macOS; Wolfram not installed in PATH	
Clean-kernel 6D	Load FIXED chain in order ?2.1; expect Gate5 ~2-3 min enum	
Clean-kernel 10D deg8	Needs keys file + planning + batches; high memory/time (45M labeled class size combinatorial)	
Python	pip install -r requirements.txt; python run_pipeline.py reproduce-6d	
Random seeds	App.A holdouts 20260804; D10 foundations seeds 20261601-20261606; Python pipeline seeds in configs	
Missing dependency	TenDDegree6ContractionPlanning_V1.wl	
External packages	None beyond Wolfram / Python reqs	

16. Current scientific conclusions

A. Established analytically

($\wedge^2$) sign; real self-duality in Lorentzian middle dimension; quadratic vanishing for chiral 5-form; (SO) vs (O) chirality exchange; even-() reduction to metrics (standard).

B. Stated in primary literature

Krull 81; Hilbert/Euler table through deg 22; uniqueness of (I₄); two deg-6; six new deg-8; twelve new deg-10 (as literature targets).

C. Reproduced exactly (project computation; MMA OBSERVED / Python partial RERUN)

6D graph counts; 6D ranks; 6D Jacobian rank 5; App. A finite holdouts; 10D component counts / ($\wedge^2$) (Python RERUN smoke).

D. Supported by modular checks

6D Jacobian primes; claimed 10D deg8 rank7 modular; Python syzygies on validation primes.

E. Supported only on finite samples

App. A identities; many 10D graph-to-basis fits; Python SVD (n_{new}).

F. Conjectural

Legacy 78/81 JSON; any claim that metric graphs already give the full (SO(1,9)) ring.

G. Unresolved

Epsilon completeness; generic stabilizer proof; deg-10 generators; full syzygies; Hironaka/HSOP; independent cross-language 10D reproduction.

17. What has not been done

- Complete reduction of all 1753 degree-8 canonical graphs onto the 7-dimensional basis
- Degree-10 graph enumeration and quotient search
- Higher-degree generators
- Systematic syzygy discovery with symbolic proof
- Generic stabilizer proof
- Complete (SO)/epsilon reduction certificate
- Complete primary/secondary classification and Hironaka decomposition
- Proof that low-degree candidates generate the full ring
- Full 81-parameter structure
- Independent cross-language 10D implementation matching Mathematica
- Formal machine-checkable proof
- Fresh-kernel retest by this auditor (Wolfram absent)

18. Recommended next steps (gated)

1. Fresh-kernel reproducibility -- Input: Research folder only. Module: driver notebook. Pass: all Gate1-5 + Deg4 V4 + Deg8 BasisValidation True in new kernel. Justifies: OBSERVED->RERUN. Unproved: 10D completeness.

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2. ($SO(1,9)$)/epsilon-reduction validation -- Explicit identities under project conventions. Pass: every tested single-() scalar equals a metric graph combination exactly. Justifies: metric-only census adequacy. Unproved: all degrees.

3. Independent exact degree-8 rank certificate -- Second implementation or exact minor export. Pass: rank7 with exhibited minor. Justifies: stronger than finite-sample alone.

4. Independent implementation (Python 10D) -- Port deg4/6 evaluations. Pass: match (I_4), two deg6, rank7. Justifies: cross-language evidence.

5. Degree-10 foundations RERUN -- Pass: product rank2 exact+modular. Justifies: target 12 conditional.

6. Degree-10 graph enumeration + quotient -- Pass: span rank14 with 12 new beyond products or documented obstruction. Justifies: computational match to Hilbert 14.

7. Syzygy discovery + holdout -- Pass: relations validate on unused primes/samples. Justifies: finite-sample identities. Unproved: symbolic.

19. Draft publication claims

Currently safer to write

- We reproduce Elamran et al. connected 3-regular counts $((1,2,6,20))$ and ranks $((1,2,3,6))$ for the Euclidean 6D 3-form, and recover generator degrees $[2,4,4,6,8]$ by product quotienting (Python numerical; Mathematica exact ranks).
- The five paper polynomials are algebraically independent (exact Jacobian rank 5 at a tested integer point).
- Appendix A relations vanish exactly on tested integer holdouts (not a symbolic proof).
- For Lorentzian self-dual 5-forms with stated conventions, $(^2=+1)$ and there are 126 real chiral components; the quadratic invariant vanishes.
- Literature targets (Cederwall): Hilbert coefficients and Krull dimension 81 are adopted as cited, not re-proved.
- Computational evidence (Mathematica, not auditor-rerun) supports dim 1,2,7 at degrees 4,6,8 for metric-only graphs, with an explicit 7-direction degree-8 basis catalogued.

Currently unsafe / overstated

- "We proved the invariant ring is generated by five elements in 6D."
- "We proved Krull dimension 81."
- "Metric-only graphs are complete for $(SO(1,9))$."
- "We found twelve degree-10 generators."
- "Degree-8 matching Hilbert 7 proves completeness."
- "78 of 81 invariants found" (docs/invariants_78.json).
- "Green TestReport => mathematical completeness."

20. Appendices

- TenD_Project_File_Manifest.csv
- TenD_Project_Claims_Register.csv
- TenD_Project_Test_Register.csv
- supporting/ -- logs, catalogs, MIT outputs, hashes, smoke rerun
- Environment: Python 3.13 venv with project requirements; Wolfram absent from PATH
- PDF engine: pure-Python text PDF writer (no TeX/pandoc installed)

End of audit report.

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